

Linear Algebra

Warm Up

* 1.

This problem involves 4×4 matrices with entries in $\mathbb{R}$. We will say that a matrix M is an ***E-matrix*** if M has 1's along the diagonal and a single nonzero entry off the diagonal. For instance:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 31 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

is an *E-matrix*. Express the matrix A below as a product of *E-matrices*.

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}$$

2.

- (a) Give an example of a finite dimensional vector space V over $\mathbb{C}$, and a nonzero linear transformation $A: V \rightarrow V$ whose only eigenvalue is 0.
- (b) Prove that the only eigenvalue of $A: V \rightarrow V$ is 0 if and only if A is nilpotent. (V is still f.d. over $\mathbb{C}$.)

3.

Give an example of a nonzero finite-dimensional vector space V over $\mathbb{C}$ and an invertible linear transformation $A: V \rightarrow V$ such that A is not diagonalizable.

Work Out

* 4.

Let M be an $n \times n$ matrix with real entries. Let M^T be its transpose. Each of these two matrices defines a linear transformation $\mathbb{R}^n \rightarrow \mathbb{R}^n$; let A denote the transformation corresponding to M , and let B denote that corresponding to M^T .

- (a) Prove that any vector in $\ker(A)$ is orthogonal to any vector in $\text{im}(B)$ with respect to the standard inner product.
- (b) Prove that $\mathbb{R}^n$ is the direct sum of $\ker(A)$ and $\text{im}(B)$.

* 5.

- (a) Let V be a finite-dimensional vector space over $\mathbb{C}$, and let $T: V \rightarrow V$ be a linear transformation. Show that T has an eigenvector.
- (b) Give an example of a nonzero finite-dimensional vector space V over $\mathbb{R}$ and a linear transformation $T: V \rightarrow V$ such that T does not have an eigenvector.
- (c) Does a linear transformation of an infinite-dimensional vector space over $\mathbb{C}$ have to have an eigenvector? Either prove this is the case, or give an example of a linear transformation of an infinite-dimensional vector space with no eigenvector.

6.

Let V and W be finite dimensional vector spaces over the same field (for the sake of argument, let's say over $\mathbb{R}$). Let $T: V \rightarrow V$ and $U: W \rightarrow W$ be linear maps. Consider the linear map $T \oplus U: V \oplus W \rightarrow V \oplus W$ which takes (v, w) to (Tv, Uw) .

- (a) Let $\mathcal{B}_1 = \{v_1, \dots, v_n\}$ be a basis for V , and $\mathcal{B}_2 = \{w_1, \dots, w_m\}$ be a basis for W . If A is the matrix of T with respect to $\mathcal{B}_1$, and B is the matrix of U with respect to $\mathcal{B}_2$, what is the matrix of $T \oplus U$ with respect to the basis $\mathcal{B} = \{(v_1, 0), \dots, (v_n, 0), (0, w_1), \dots, (0, w_m)\}$?
- (b) What are the eigenvalues of $T \oplus U$ in terms of the eigenvalues of T and the eigenvalues of U ?
- (c) What is the trace of $T \oplus U$ in terms of the trace of T and the trace of U ?
- (d) Same question for the determinant of $T \oplus U$.
- (e) Same question for the characteristic polynomial of $T \oplus U$.
- (f) Let $\mu_T(x)$ be the minimal polynomial of T , and $\mu_U(x)$ be the minimal polynomial of U . Prove that the minimal polynomial of $T \oplus U$ is $\text{lcm}(\mu_T, \mu_U)$.

* 7.

Suppose V is a finite dimensional vector space over $\mathbb{C}$, and $T: V \rightarrow V$ is a linear operator. Suppose the subspace of V spanned by all eigenvectors of T is 1-dimensional. Prove that the minimal polynomial of T is $(x - \alpha)^n$, where $n = \dim V$ and α is some complex number.

8.

- (a) Give an example of two linear transformations with the same characteristic polynomial but different minimal polynomials.
- (b) Give an example of two linear transformations with the same minimal polynomials which are not similar.

* 9.

(Originally part (d) of problem 5.) Suppose that T and U are two linear transformations on a finite-dimensional vector space V over $\mathbb{C}$ which satisfy $TU = UT$. Prove that there is some $v \in V$ which is an eigenvector for both T and U .